

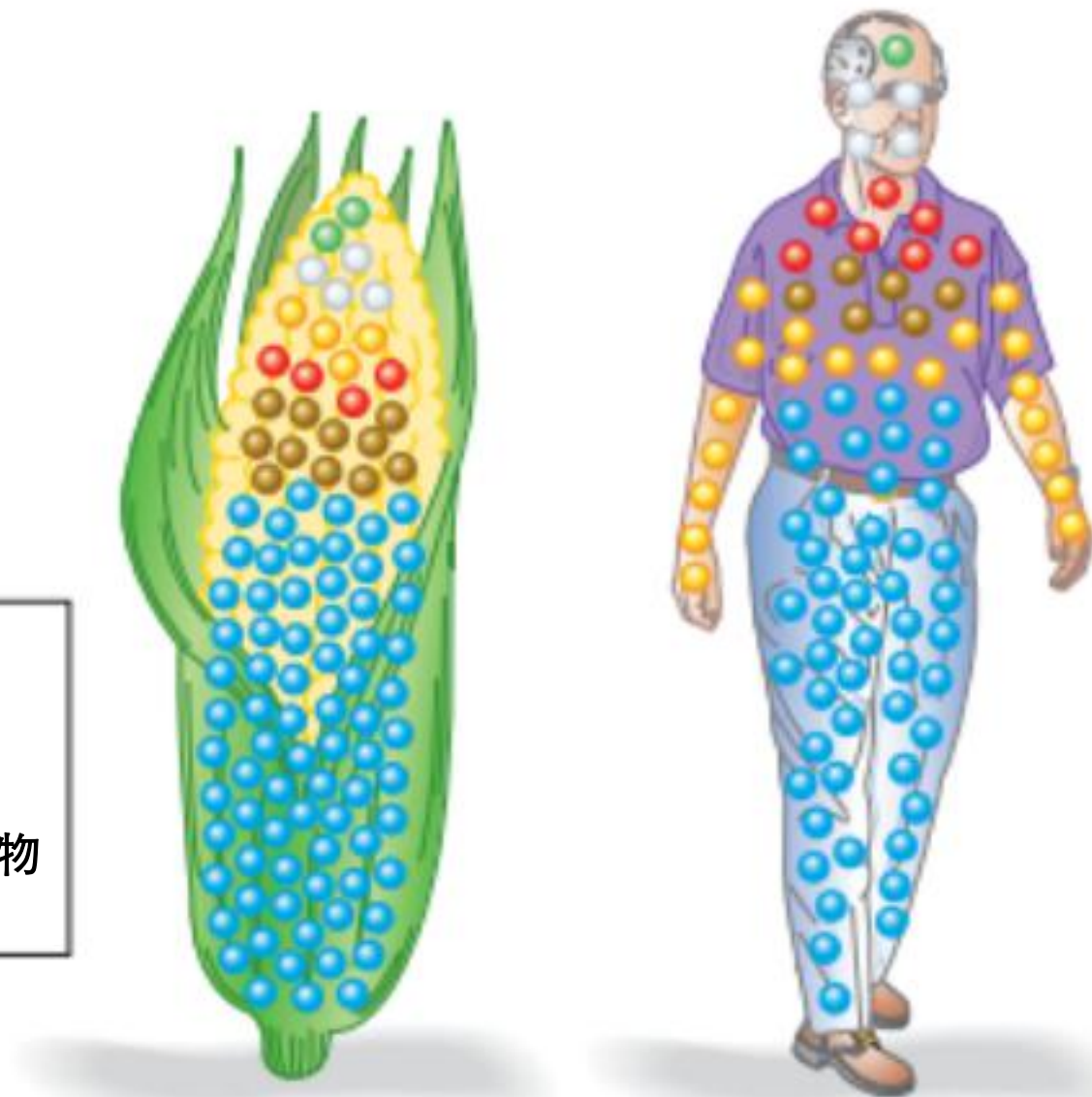
身体与食物

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• 一）、身体的组成成分

Foods and the human body are made of the same materials.



二、身体与食物

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	碳	氧	氢	氮	矿物质
水		✓	✓		
碳水化合物	✓	✓	✓		
脂肪	✓	✓	✓		
蛋白质	✓	✓	✓	✓	#
维生素	✓	✓	✓	✓ *	#
矿物质		✓	✓		

* B族维生素含氮； # 蛋白质和一些维生素含硫； 维生素B12含矿物质钴。

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- 二）、必需营养素

- 机体需要，但不能自身合成，只能通过食物获取的营养素。包括：水、部分脂肪酸（脂肪）、部分碳水化合物（膳食纤维）、部分氨基酸（蛋白质）、几乎所有的维生素和所有的矿物质。

- 三）能量

- 脂肪： 9千卡/克
- 碳水化合物： 4千卡/克
- 蛋白质： 4千卡/克
- 酒精： 7千卡/克

二、身体与食物

- 三)、能否依靠膳食补充剂?
- 医学上:
 - ✓ 膳食种类: 基本膳食、治疗膳食、诊断膳食、代谢膳食等。
 - ✓ 营养支持途径: 肠内营养、肠外营养。
- 肠外营养: 上海周绮思
- 肠内营养: Soylent: 快捷、方便、科学、时髦、硅谷光环等

• 三)、能否依靠膳食补充剂?

Version	Changes	Date
Soylent 1.0	First full version. Ingredients were finalized in January 2014, which use rice as the protein source ^[22] and shipments began in April ^[23] (vegan) and May ^[24] (regular) of 2014.	Early 2014
Soylent 1.1	The sucralose was decreased, giving it a more neutral flavor, and new digestive enzymes were added. ^[25]	October 2, 2014 ^[26]
Soylent 1.2	Omega-3 fatty acid from fish sources was replaced with omega-3 from algae , making the drink suitable for vegans and the enzymes added in Soylent 1.1 were removed. ^[17]	November 10, 2014 ^[17]
Soylent 1.3	Dipotassium phosphate was added and shipping box sizes were reduced. ^[27]	December 11, 2014 ^[27]
Soylent 1.4	Fats were incorporated into the powder which eliminated the need for the oil bottles, resulting in less packaging required in the shipping boxes and isomaltulose was added. Gum acacia was removed. ^[28]	February 25, 2015 ^[29]
Soylent 1.5	Improvements to texture from a reduction to oat flour and an addition of emulsifiers. Removal of powdered safflower and flaxseed oil which were both replaced by canola oil powder, supplementing the existing powdered high oleic sunflower oil and algal oil. ^[19]	June 1, 2015 ^[19]
Soylent 2.0	First pre-mixed Soylent liquid product and it alters the fat/carb/protein ratios to 47/33/20; it has a glycemic index of 49.2 and a glycemic load of 16.7. ^[20] About half of the lipid calories come from algal sources and it uses soy for its protein source. ^[21]	September 9, 2015 ^[30]

• 三）、能否依靠膳食补充剂？

Soylent 2.0 (Drink Version) Nutrition Facts

Serving Size 1 bottle Soylent (414 mL)	Servings per Container: 1	
Calories	400	
Calories from fat	190	
	Amount per Serving	% Daily Value*
Total Fat	21g	35%
Saturated Fat	2g	10%
Trans Fat	0g	N/A
Cholesterol	0 mg	0%
Sodium	300 mg	13%
Potassium	700 mg	20%
Total Carbohydrate	37g	12%
Dietary Fiber	3g	12%
Sugars	9g	
Protein	20g	

二、身体与食物

- 四）、食物的功能

- 1.食管和胃肠道是有生命的器官、牙齿；
- 2.食物提供营养素，也提供情感上、心理上的满足；
- 3.社交的需要；
- 4.食物富含生物活性物质。

小结

- ✓ 食物提供人体所需的各种营养素—水、蛋白质、碳水化合物、脂肪、维生素和矿物质；
- ✓ 产能营养物质：碳水化合物（4千卡/克）、脂肪（9千卡/克）、蛋白质（4千卡/克）；酒精也提供能量（7千卡/克）；
- ✓ 除了提供营养素，食物还满足心理和社交需要、促进消化器官活动、提供食物活性物质；